

Inferring prior distributions of recruitment compensation metrics from life-history parameters and allometries

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Abstract: The recruitment compensation metrics, the stock–recruit steepness (h) and compensation ratio (κ), are difficult to estimate, yet they are integral components of contemporary fishery models. To aid in the estimation of κ and h , a hybrid method to construct their prior distributions for a species is developed. The method is hybrid because it integrates (i) a meta-analysis of the relationship between maximum rates of recruit production (α) and asymptotic lengths obtained across fish species of different life histories and (ii) species-specific unfished spawning biomass per recruit (Φ_0). This method is applied to five finfish species found off the East Coast of the United States. Uncertainty is introduced by sampling growth parameters, α , and natural mortality and — through Monte Carlo simulations — propagated into Φ_0 , κ , and h for the Beverton–Holt and Ricker stock–recruit relationships. Descriptive statistics and parameters from probability density functions of the simulated distributions of κ and h are generated. The method developed requires fewer inputs than the reproductive ecology method while likewise allowing the development of species-specific statistics for κ and h . These statistics, rather than their counterparts generated from classical meta-analyses, are better suited for use in fishery models.

Résumé : Les paramètres de recrutement compensatoire, à savoir la pente de la relation stock–recrues (h) et le rapport de compensation (κ) sont difficiles à estimer, bien qu'ils soient des composantes essentielles des modèles halieutiques contemporains. Pour faciliter l'estimation de κ et de h , une méthode hybride pour en construire des distributions a priori pour une espèce est mise au point. La méthode est dite hybride parce qu'elle intègre (i) une métaanalyse d'une relation entre les taux maximums de production de recrues (α) et les longueurs asymptotiques obtenues pour différentes espèces de poissons ayant différents cycles biologiques et traits démographiques, et (ii) pour une espèce donnée, la biomasse féconde non exploitée par recrue (Φ_0). Cette méthode est appliquée à cinq espèces de poissons à nageoires des eaux américaines de l'océan Atlantique. Une incertitude est introduite en échantillonnant des paramètres de croissance, α , et la mortalité naturelle et, à l'aide des simulations de Monte Carlo, elle est propagée dans Φ_0 , κ , et h pour les relations stock–recrues de Beverton–Holt et Ricker. Des statistiques descriptives et paramètres tirés de fonctions de densité de probabilité des distributions simulées de κ et h sont générés. La méthode mise au point nécessite moins de données d'entrée que la méthode basée sur l'écologie de la reproduction tout en permettant le développement de statistiques en question pour κ et h propres à une espèce donnée. Les statistiques ainsi générées devraient être utilisées dans les modèles halieutiques, plutôt que leurs équivalents découlant de métaanalyses classiques. [Traduit par la Rédaction]

Introduction

In the past two decades, parametric stock–recruit relationships (SRRs) have become integral modules of statistical catch-at-age (SCAA) and statistical catch-at-stage (SCAS) models (e.g., Walters et al. 2006; Methot and Wetzel 2013; Carruthers et al. 2014). SCAA and SCAS models project exploited populations forward in time, directly estimating recruitment and calculating biological reference points that account for the relationship between spawners and resulting recruits, thereby allowing maximum sustainable yield (MSY) to be determined (see Shepherd 1982 for age-structured dynamic pool models and SRRs; see Munyandorero 2015 for stage-structured dynamic pool models and SRRs). This procedure contrasts with sequential population methods (i.e., virtual population analysis) in which recruitment is derived using backward recursion (instead of being estimated based on an SRR). This latter method requires that SRRs and biological reference points associated with maximum sustainable yield be developed externally

based on the stock assessment model outputs of spawners and recruits (e.g., Gavaris 1988). SRRs used in SCAA and SCAS models have been reparameterized to have meaningful biological interpretations based on steepness (Mace and Doonan 1988) or the Goodyear recruitment compensation ratio (Goodyear 1977, 1980, 1993), which is hereinafter referred to as compensation ratio. In reparameterized SRRs, steepness is the ratio of the recruitment produced by 20% of the unfished parental stock to the unfished recruitment (Francis 1992). The compensation ratio is the ratio of (i) egg-to-recruit survival for the very low number of total eggs produced (i.e., the number of total eggs produced approaches zero) to (ii) egg-to-recruit survival for the number of unfished total eggs (Walters and Martell 2004; van Poorten et al. 2018).

Compensation ratio and steepness are related following a 1:1 nonlinear increasing function and therefore are interchangeable (e.g., Martell et al. 2008). Demographically, compensation ratio and steepness are alternative measures of the degree of density dependence (compensation) in recruitment to the parental stock;

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for this reason, they are herein together referred to as recruitment compensation metrics. Steepness is employed more often than is the compensation ratio, but regardless of which parameter is used, it is difficult to estimate, rendering uncertain the achievability of incorporating SRRs into SCAA and SCAS models. The difficulty of estimating recruitment compensation metrics arises from, for instance, uninformative fishing history. For example, insufficient fishery data often result in a time series of spawning-stock biomass and recruitment that lacks the necessary statistical contrast to estimate recruitment levels at low or high stock sizes (Walters and Martell 2004; Magnusson and Hilborn 2007; Conn et al. 2010; Magnusson 2016). Thus, in analyses using the Beverton–Holt SRR (BH-SRR), in which steepness is constrained between 0.2 and 1, steepness tends to be estimated near the lower or upper bounds (Conn et al. 2010; Lee et al. 2012). Difficulties in estimating steepness are one reason that many studies have been devoted to constructing prior distributions of steepness, as it helps to reduce parameter confounding in SCAA and SCAS models (Brodziak et al. 2015 and references therein). But since steepness and the compensation ratio are functionally related nonlinearly and positively and can be interconverted (Martell et al. 2008; Hillary et al. 2012; Shertzer and Conn 2012), a prior distribution obtained for steepness is implicitly reflected in the compensation ratio. In this study, the prior distributions of both recruitment compensation metrics are explicitly developed; this may be particularly useful for the compensation ratio, because its use would be more suitable than that of steepness according to Walters et al. (2006), Hillary et al. (2012), and Brooks (2013).

There are two main strategies to ease the estimation of steepness in SCAA and SCAS models. The first fixes steepness at a value regarded as reasonable or a best guess. The second strategy consists of constructing prior distributions (henceforth “priors”) of steepness, which are used as penalty functions in a maximum-likelihood context or to inform Bayesian applications. Steepness priors are commonly constructed through meta-analyses (e.g., Myers et al. 2002; Shertzer and Conn 2012; Punt and Dorn 2014) and reproductive ecology analyses (Mangel et al. 2010; Simon et al. 2012; Brodziak et al. 2015; Cortés 2016; Kai and Fujinami 2018).

The concept of reproductive ecology for developing steepness priors was introduced by Mangel et al. (2010) to account for evolutionary ecology and reproductive biology drivers (e.g., reproductive timing, spatial and temporal life cycles, larval dispersion) of steepness. Contrary to meta-analyses, which employ maximum rates of recruit production based on SRR curves of several species, reproductive ecology analyses calculate the maximum rate of recruit production of a single species as the ratio of the simulated number of recruits to the simulated biomass of spawners. In these calculations, the expected number of recruits is equal to the simulated total egg production multiplied by egg-to-recruit survival for a low spawning-stock biomass, where egg-to-recruit survival accounts for the durations and growth rates of early life stages. Like meta-analyses, reproductive ecology analyses incorporate the unfished spawning biomass per recruit (SBPR) to develop steepness priors. For both methods, the unfished SBPR involves life-history variables of the recruited age groups (growth and natural mortality rates, proportions of mature individuals, sex ratios). However, the reproductive ecology method attempts to include as many reproductive characteristics as possible, such as batch fecundity and frequency and spawning season duration. Meta-analyses generate steepness priors assuming that these priors are applicable to a collection of species believed to have similar life histories. Reproductive ecology methods generate steepness priors for a single species by combining the unfished SBPR and maximum rate of recruit production based on growth and survival rates of prerecruit stages.

In estimating egg-to-recruit survival of finfish species, reproductive ecology applications have so far relied on egg-to-recruit total mortalities inferred from meta-analytic relationships be-

tween early-stage natural mortality and dry weight (Mangel et al. 2010; Simon et al. 2012; Brodziak et al. 2015). Thus, those applications have not yet managed to incorporate behavioral ecology factors (sensu Goodyear 1980; Dill 2017; Lowerre-Barbieri et al. 2017; van Poorten et al. 2018) that, at the species level, drive egg-to-recruit survival. Dill (2017) defines behavioral ecology as “an evolutionary-based discipline that attempts to predict how animals will behave in a given set of environmental circumstances and how those behavioural decisions will impact population growth”. For early life stages of a species, behavioral ecology factors include selection of nursery habitats, competition for food and spawning sites, predation, cannibalism, reproductive timing, parental care or not, stressful or favorable environmental and weather conditions, and degradation or restoration of nursery habitat by humans. Those factors induce both density-dependent and density-independent mortalities (Goodyear 1980), which are not easy to estimate for so many exploited populations. Therefore, until behavioral ecology factors are analyzed, and their induced mortalities quantified for integration in recruitment compensation metrics of individual finfish species, the metrics in question will likely continue to be based solely on meta-analytic relationships and life-history parameters (including reproductive traits), especially those of recruited stages. In other words, life-history parameters of the recruited stages are often the only data available to determine the recruitment compensation metrics owing to the paucity of early-stage natural mortality rates induced by behavioral ecology factors.

The objective of the present study is to develop a method for constructing priors of recruitment compensation metrics for a species by combining facets of meta-analyses and reproductive ecology. The methodology in question relies on three types of information: (i) a metadata of density-independent maximum rates of recruit production (i.e., slopes at the origins of SRR curves, expressed as number of recruits per unit of spawning-stock biomass) obtained across fish species irrespective of life-history similarity; (ii) asymptotic lengths of fish species for which the aforementioned maximum rates of recruit production are compiled; and (iii) parameters from the literature on a species’ life history (e.g., growth parameters and longevity, weight-length scales and exponents, female sex ratio by age, proportion of mature females by age, and monthly proportions of spawning-capable females). The first step is to develop an empirical relationship between maximum rates of recruit production and asymptotic lengths. Next, this relationship is used to infer a plausible maximum rate of recruit production for a species, given a random asymptotic length. Third, the life-history parameters of a species are used to calculate the unfished SBPR, which involves a statistic accounting for the protracted spawning nature of fish populations. Finally, the inferred maximum rate of recruit production and the unfished SBPR are combined to calculate and construct the distributions of the recruitment compensation metrics.

Specifically, this study aims to show that recruitment compensation is strongest in long-lived species, a property that was suggested in previous studies (e.g., Myers et al. 2002; Goodwin et al. 2006; Andersen and Beyer 2015; Brodziak et al. 2015). To achieve this objective, the method developed in this study is applied to five finfish species found off the East Coast of the United States. The following species were selected mainly for their contrasting life-history traits, specifically size and longevity, as well as their economic value: Atlantic croaker (*Micropogonias undulatus*), a medium-bodied and medium-lived fish; bluefish (*Pomatomus saltatrix*), a large-bodied and medium-lived fish; red snapper (*Lutjanus campechanus*), a large-bodied and long-lived fish; sheepshead (*Archosargus probatocephalus*), a medium-bodied and long-lived fish; and spot (*Leiostomus xanthurus*), a small-bodied and short-lived fish.

Table 1. Notations used for constructing priors of the recruitment compensation metrics.

Symbol	Description
h	Stock–recruit steepness (unitless)
κ	Goodyear recruitment compensation ratio (unitless)
α	Slope at the origin of an SRR curve — used along with the parameter β for both the Beverton–Holt and Ricker stock–recruit relationships and has the unit of recruit per spawning biomass. Therefore, consistent with Denney et al. (2002) and Goodwin et al. (2006), α corresponds to maximum rate of recruit production.
L_{∞}	Asymptotic length for the von Bertalanffy growth function (VBGF; cm or mm)
K	Growth rate for the VBGF (year^{-1})
t_0	Theoretical age at length 0 for the VBGF (year)
a	Weight–length scale
b	Weight–length exponent
L_t	Mean length at age t (years), based on the VBGF (cm or mm)
W_{∞}	Asymptotic weight for the VBGF; $W_{\infty} = aL_{\infty}^b$ (kg)
W_t	Weight at age based on the VBGF (kg)
ρ_t	Sex ratio at age
M	Constant natural mortality (year^{-1})
L_r	Reference length (cm or mm)
M_r	Natural mortality (year^{-1}) at L_r (cm or mm)
$L_{50\%}$	Length at 50% maturity (cm or mm)
S_t	Natural survival at age (year^{-1})
M_t	Natural mortality at age (year^{-1})
l_t	Survivorship (probability of surviving) to age t in the absence of fishing (year^{-1})
μ_t	Proportion of mature females at age
π_m	Proportion of spawning-capable females in month m
φ_m	Spawning offset (fraction of the year representing the time between the start of the year and the beginning of month m , during which spawning does or does not take place); φ_m is calculated assuming that M_t is uniformly distributed over the year. By convention, φ_1 for January is zero, $\varphi_2 = 1/12$ for February, consistent with the elapsed month of January, and so on.
λ_t	Age-specific weighted mean of monthly survival rates
$T_{\max}$	Maximum age (years)
Φ_0	The unfished spawning biomass per recruit (kg)

Methods

Basic relationships and metadata

For a single-species stock, the construction of a prior for the compensation ratio, κ , and steepness, h (see Table 1 for all notations used), relies on the following allometry, definition, and function. The allometry consists of a negative and significant relationship between the SRR parameter α (maximum rate of recruit production) and the parameter L_{∞} of the von Bertalanffy growth function, VBGF (Denney et al. 2002; Goodwin et al. 2006; Hall et al. 2006; Munyandorero 2018). In this context, if a species' L_{∞} value is in the domain of the equation describing an empirical relationship between α and L_{∞} , a value of α can be inferred (Hall et al. 2006; Rochet et al. 2011; Thorpe et al. 2015; Munyandorero 2018). The definition consists of expressing κ as α multiplied by the unfished SBPR Φ_0 (e.g., Goodyear 1980; Walters and Martell 2004). Finally, the equation relating h to κ is a nonlinear 1:1 increasing function (e.g., Martell et al. 2008). Figure 1 shows how h and κ vary with each other, based on the equations of h and κ in Martell et al. (2008).

In developing an empirical relationship between α and L_{∞} , estimates of α (the number of recruits per metric tonne of spawning-stock biomass) and L_{∞} (cm) for exploited fish populations of different life histories are compiled based on the criterion that α values have been obtained using stock–recruit time series of spawning-stock biomass and recruits generated through sequential population models (i.e., virtual population analysis). The α values have been taken from the literature, mainly from Goodwin et al. (2006), because the stock assessment methods and outputs of spawning-stock biomass and recruits were comparable to each other. Admittedly, such stock assessment model outputs should not be treated as data (Brooks and Deroba 2015), but these outputs provided the best reconstruction of nature and were used for management decisions. Preferably, values of α and L_{∞} come from the same study. When values of only α are available, values of L_{∞} are obtained elsewhere in the literature, including FishBase

(<http://fishbase.org>), a major global online database of fish parameters. The compiled estimates of α and L_{∞} ($n = 68$) are jointly treated as metadata.

Values of α are compiled irrespective of the form of the SRR from which they were derived because the parameter α retains the same definition for all SRRs (Shepherd 1982; Johnston et al. 2016; see Brooks and Powers 2007 on analytical demonstrations for the BH-SRR and the Ricker SRR (R-SRR); note that the Shepherd SRR (S-SRR) simply generalizes the BH-SRR by adding a shape parameter to the density-dependent term. But the BH-SRR must be of the form $R = \alpha S / (1 + \beta S)$). Moreover, in studies where α values have been estimated for different SRRs, the BH-SRRs are preferred. The reason is that the BH-SRR is the most common choice in analyses, and the BH-SRR should be the default unless there are mechanisms, such as cannibalism, that favor the R-SRR (Shertzer and Conn 2012) or the S-SRR (Shepherd 1982).

An α – L_{∞} empirical relationship

Following Hall et al. (2006), a linear model (eq. 1) is fitted to the log-transformed values of α and L_{∞} in the aforementioned metadata:

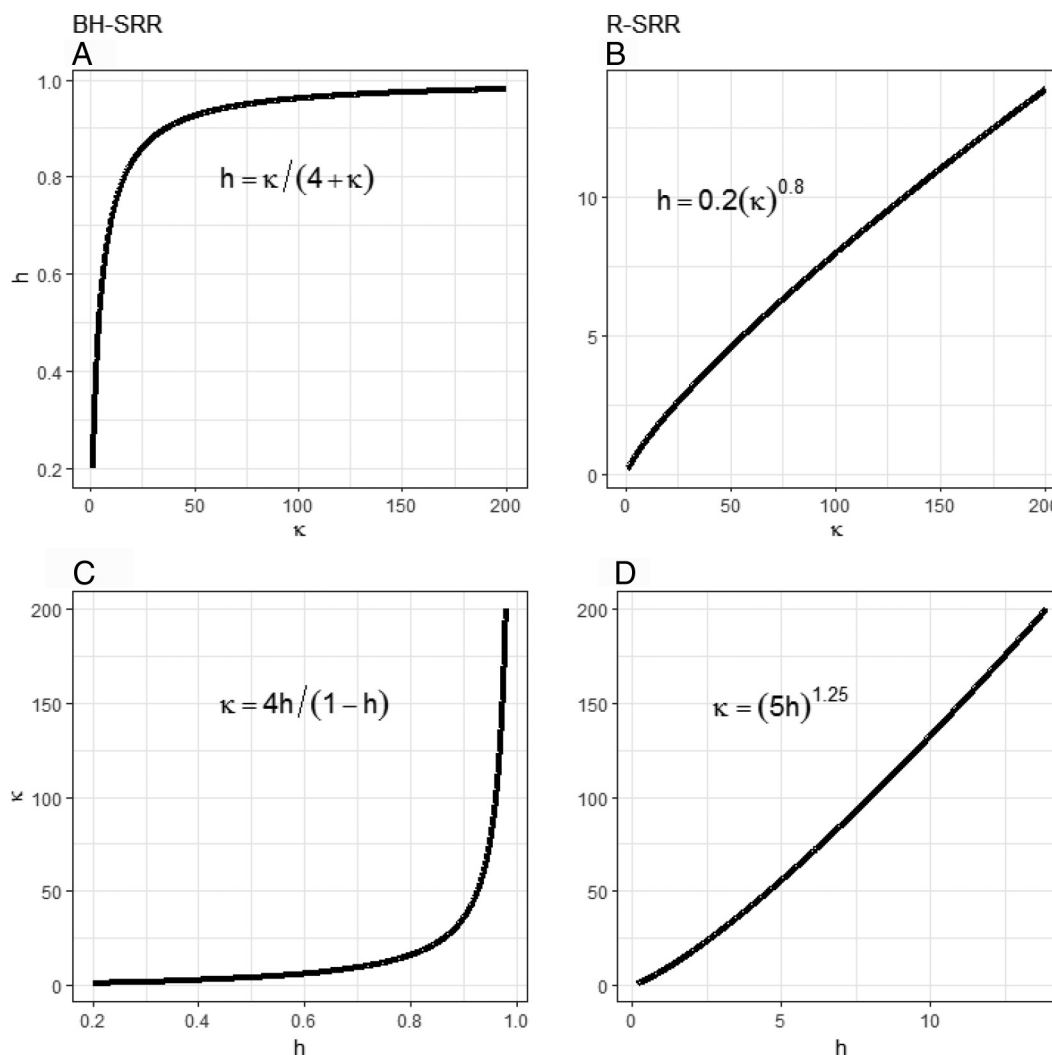
$$(1) \quad \ln \alpha = j + k \ln L_{\infty}$$

where j and k are the intercept and slope, respectively.

Life-history inputs to the unfished SBPR

The unfished SBPR, Φ_0 , is calculated using age-specific natural mortality, M_t (calculated below); mean weight, W_t (calculated below); sex ratio, ρ_t (i.e., the proportion of females in a sample composed of individuals of both sexes); proportion of mature females, μ_t (i.e., the ratio of predefined (e.g., only ripe-running or both developing and ripe-running) stages of mature females to the total number of females for which gonad development was studied); the monthly proportion of spawning-capable females, π_m ; and the

Fig. 1. Variations of (A) steepness (h) against the compensation ratio (κ) for the Beverton–Holt stock–recruit relationship (BH-SRR), (B) h against κ for the Ricker stock–recruit relationship (R-SRR), (C) κ against h for the BH-SRR, and (D) κ against h for the R-SRR (source of equations used for plotting: Martell et al. 2008).



monthly spawning offset, φ_m (see Table 1 for complete definitions). The proportion of spawning-capable females π_m is the fraction of females which, in each month of a year, have large ovaries, prominent blood vessels, and visible individual oocytes, and histologically show features like vitellogenic oocytes or evidence of imminent spawning (e.g., hydration and ovulation; see Brown-Peterson et al. 2011 and Lowerre-Barbieri et al. 2011 for details). The spawning offset φ_m is the fraction of the year representing the time between the start of the year and the beginning of month m (during which spawning does or does not take place or, equivalently, π_m is or is not equal to zero). The spawning offset is calculated assuming that natural mortality (M_t) is uniformly distributed over the year. By convention, φ_1 for the month of January is zero; $\varphi_2 = 1/12$ for the month of February, consistent with the elapsed month of January; and so on, such that, for the month of December, $\varphi_{12} = 11/12$. Note that, in the above schedules by age, the subscript t denotes age in years. Note also that all proportions are species specific except φ_m .

Now, consider a deterministic analysis or a realization of stochastic simulations of the unfished SBPR (Φ_0) of a species, where calculations involve one set of VBGF parameters (L_∞ , K , and t_0), one set of weight–length scale and exponent (a and b), and one schedule for each of the reproductive proportions ρ_t , μ_t , π_m , and φ_m . The sequence of equations presented below describes fish growth in

size and mass, natural mortalities, survivals, and survivorships derived using the aforementioned growth parameters and reproductive proportions. It is intended to represent the order of calculations leading to Φ_0 and, ultimately, to κ and h , given α inferred from eq. 1 (the proportions ρ_t and μ_t are directly integrated in the computation of Φ_0 and, like other reproductive proportions, are not herein described by equations). Unnumbered equations are not required in the calculations but are worth showing because they underlie the numbered equations following them.

Natural mortality by age (M_t) depends on two equations. First is an estimator of constant natural mortality, M , which is Pauly's nonlinear empirical equation (Then et al. 2015, 2018):

$$(2a) \quad M = 4.118K^{0.73}L_\infty^{-0.33} \text{ (if } L_\infty \text{ in cm) or}$$

$$M = 8.87K^{0.73}L_\infty^{-0.33} \text{ (if } L_\infty \text{ in mm)}$$

Second is the standard VBGF:

$$(2b) \quad L_t = L_\infty[1 - e^{-K(t-t_0)}]$$

where L_∞ is asymptotic length, K is growth rate, t_0 is the theoretical age at length 0, and L_t is mean length at age.

Natural mortality by age (M_t) follows from Lorenzen's (2000, 2005) natural survival equation, S_t , based on the VBGF:

$$S_t = \left[\frac{L_t}{L_t + L_\infty(e^K - 1)} \right]^{\frac{M_r}{KL_\infty}}$$

where M_r is the natural mortality at a conveniently chosen reference length L_r , which is herein predicted by the VBGF; L_r has the same units as L_∞ (cm or mm).

Following Lorenzen's (2000, 2005, 2011) notations of M_t (e.g., M_l , $M(l)$, or $M(L)$), M_t is natural mortality at mean body length L_t at age t . According to Lorenzen (2011), M_t should be expressed as a function of L_t relative to L_r such that natural mortality at the age associated with L_r matches M . Under this condition, it follows from the previous natural survival equation that

$$M_r = \frac{MKL_\infty}{\ln \left[1 + \frac{L_\infty}{L_r}(e^K - 1) \right]}$$

and that

$$(2c) \quad M_t = M \times \frac{\ln \left[1 + \frac{L_\infty}{L_t}(e^K - 1) \right]}{\ln \left[1 + \frac{L_\infty}{L_r}(e^K - 1) \right]}$$

The denominator of eq. 2c is a constant; for a growth trajectory characterized by L_∞ and K , M is unique (eq. 2a). These two aspects influence the levels of M_t and survivorship l_t (i.e., the probability of surviving to age t in the absence of fishing) in response to alternative choices of L_r . In fact, lowering L_r increases the denominator of eq. 2c and results in smaller values of M_t and greater values of l_t ; an increase of L_r produces a higher M_t and reduces l_t . Note also that the procedure of matching M to natural mortality at the age related to L_r differs from a technique that treats M as target M , thereby rescaling M_t such that the mean of M_t values from the age associated with L_r to a specified maximum age equals M (Clay 2011; Kai and Fujinami 2018).

Two age-specific schedules follow from the calculated natural mortality by age M_t (eq. 2c). The first schedule is survivorship l_t , as defined above:

$$(2d) \quad l_t = 1 \text{ for age } 0 \text{ and } l_t = l_{t-1} \exp(-M_{t-1}) \text{ for age } > 0$$

The second schedule — the only quantity involving the monthly proportions of spawning-capable females (π_m) and spawning off-sets (φ_m) — is the statistic λ_t :

$$(2e) \quad \lambda_t = \frac{\sum_{m=1}^{12} \pi_m e^{(-\varphi_m M_t)}}{\sum_{m=1}^{12} \pi_m}$$

The statistic λ_t clearly is the weighted mean of monthly survival rates, $e^{(-\varphi_m M_t)}$, for t -years-old fish alive on 1 January that still survive at the beginning of any month of a year, but spawn only in a single month; the weights of $e^{(-\varphi_m M_t)}$ are the proportions π_m (Munyandorero 2018). Note that for the month of January, $e^{(-\varphi_m M_t)} = 1$, because $\varphi_m = \varphi_1 = 0$. The statistic λ_t is in line with equations by

Gabriel et al. (1989) and Haddon (2001) for calculating the unfished SBPR for fish alive on 1 January that still survive at the beginning of the month in which the spawning season peaks (e.g., if the spawning season peaks in July, fish alive on 1 January are, according to these authors, decremented by $e^{(-0.5M_t)}$). Instead of restricting the unfished SBPR analysis to a month in which spawning peaks, the statistic λ_t is intended to account for a protracted spawning season of fish populations in the calculation of the unfished SBPR.

The final equation in the calculation of Φ_0 relates to mean weight at age W_t (kg):

$$(3) \quad W_t = W_\infty [1 - e^{-K(t-t_0)^b}]^b$$

where K is growth rate, t_0 is the theoretical age at length 0, and W_∞ (kg) is asymptotic weight ($W_\infty = aL_\infty^b$) of the VBGF; L_∞ is asymptotic length; and a is the scale and b the exponent of the weight-length relationship.

Calculating the unfished SBPR

The age-specific unfished SBPR (kg) for females starting from age-0, Ψ_t , is given by

$$(4) \quad \Psi_t = l_t \rho_t W_t \mu_t \lambda_t$$

The age-aggregated unfished SBPR (kg) is given by

$$(5) \quad \Phi_0 = \sum_{t=0}^{T_{\max}} \Psi_t$$

where $T_{\max}$ is the maximum observed age, assumed to represent the species' longevity.

Recruitment compensation metrics

The metrics describing the compensatory recruitment are the compensation ratio, κ , and steepness, h . According to Walters and Martell (2004), κ is given by

$$(6) \quad \kappa = \alpha \Phi_0$$

In other words, κ is the ratio of the maximum rate of recruit production to the rate of recruit production at the unfished spawning biomass ($1/\Phi_0$). Equivalently, κ is a constant of proportionality between the slope at the origin of an SRR curve and the slope of the unfished replacement line of that curve. The value of κ must be >1 for a population to persist, even in the absence of fishing (Martell et al. 2008; Mangel et al. 2010).

The metric h follows from κ (Brooks and Powers 2007) and is herein derived for the BH-SRR and R-SRR, because these SRRs are commonly used in stock assessments and simulations. For the BH-SRR, h results from the equation $R = \alpha S / (1 + \beta S)$, where R is recruitment, S is spawning-stock biomass, and α and β are parameters:

$$(7a) \quad h = \frac{\kappa}{4 + \kappa}$$

For density-dependent recruitment to occur through the BH-SRR, h must be >0.2 and <1 , because in eq. 7a, $\kappa > 1$ and the numerator is smaller than the denominator. Analyses using $h = 1$ automatically cancel out the parental stocks in BH-SRRs; in a deterministic context, this scenario results in recruitment being equal to the so-called unfished recruitment.

The above form of the BH-SRR is chosen to be consistent with studies that used eq. 7a to calculate h (e.g., Myers et al. 2002;

Brooks and Powers 2007; Martell et al. 2008). If h were based on the BH-SRRs of the forms $R = \alpha S/(\beta + S)$ and $R = S/(\alpha + \beta S)$, it would be calculated as, respectively, $h = \kappa/(4\beta + \kappa)$ (Cervino et al. 2013) and $h = \Phi_0/(4\alpha + \Phi_0)$. Note, however, that the parameters α and β for various forms of the BH-SRR are not equal.

For the R-SRR, $R = \alpha S \exp(-\beta S)$, and h is calculated as

$$(7b) \quad h = 0.2(\kappa)^{0.8}$$

The range of h for the R-SRR is $(0.2, \infty)$.

Characterizing uncertainty and stochasticity

Random draws deal solely with the growth parameters (a , b , L_∞ , K , and t_0), maximum rate of recruit production (α), and natural mortality (M), and are performed using the program R version 3.4.3 (R Core Team 2018). Other quantities (length at age, natural mortality by age, survivorship to an age, the statistic λ_t , asymptotic weight and mean weight at age, the unfished SBPR, and recruitment compensation metrics) are stochastically derived, also using R, based on random samples of growth parameters, α , and M (see Supplementary materials ¹ for details).

For a focal species, consider the availability of several weight-length scales (a) and exponents (b) and of several VBGF parameters (L_∞ , K , and t_0) obtained from the literature. The number of pairs (a , b) and the number of trios (L_∞ , K , t_0) are not necessarily equal. These growth parameters are herein treated as observed data.

The variability in the estimation of a and b and of L_∞ , K , and t_0 are assumed to reflect scientific uncertainty. Further, once the observed growth data are deemed appropriate for an analysis (especially if they have been applied elsewhere), no unique pair (a , b) and no unique trio (L_∞ , K , t_0) are a priori considered better than other such pairs and trios for proper analyses of the unfished SBPR. In other words, all pairs (a , b) on the one hand and all trios (L_∞ , K , t_0) on the other are assumed to be independent of each other and to be equally reliable in SBPR calculations.

To characterize uncertainty in growth parameters, the observed values of a and b on the one hand and of L_∞ , K , and t_0 on the other are jointly resampled first with replacement (number of replicates $r = 10\,000$), using the sample function in R conditional on their means; this sampling procedure is necessary and may be enough to characterize uncertainty in growth parameters. Since the resampled values of the previous parameters are approximately normally distributed, given their simulated mean vectors and variance-covariance matrices (calculated with the cov function in R), a and b are then drawn as bivariate normal distributions, and L_∞ , K , and t_0 as trivariate normal distributions. Sampling is performed ($r = 10\,000$) with the mvrnorm function in the R package MASS (Venables and Ripley 2002). Appendix A gives details on the sampling procedures and sampling distribution errors used to incorporate parameter uncertainty.

The maximum rates of recruit production α are sampled from a log-normal distribution error with a standard deviation (SD) of 0.908 and a mean calculated using eq. 1 (Appendix A). The value of 0.908 is the SD of the residuals of the fitted regression for eq. 1 and so is on a log scale. Uncertainty in constant M is introduced by sampling M values from a log-normal distribution error with a coefficient of variation (CV) of 0.5 and a mean obtained with eq. 2a on log scale (Appendix A). A CV of 0.5 was suggested by MacCall (2009) as a minimal default value for the variability of M on a log scale.

Given random realizations of a , b , L_∞ , K , t_0 , α , and M , 10 000 Monte Carlo simulations are run on eq. 2b through eq. 7b (Supplementary materials ¹). For each sequence of simulations, a set of random values of a , b , L_∞ , K , t_0 , α , and M is first used to derive

successively (i) mean length at age L_t (eq. 2b), (ii) natural mortality by age M_t (eq. 2c), (iii) survivorship to age l_t (eq. 2d), (iv) the statistic λ_t (eq. 2e), and (v) asymptotic weight W_∞ and mean weight at age W_t (eq. 3). Next, the values of l_t , W_t , and λ_t are combined with fixed values of the sex ratio by age (ρ_t) and proportion of mature females by age (μ_t) to calculate the unfished SBPR Φ_0 (eqs. 4 and 5). Last, the unfished SBPR Φ_0 is multiplied by the maximum rate of recruit production (α) to derive the compensation ratio κ from eq. 6, which in turn is used to calculate steepness h (eqs. 7a and 7b). This way, the 10 000 sequences of simulations generate different random values of Φ_0 , κ , and h , which are the basis for constructing and describing the priors of these parameters. For h calculations in the simulated BH-SRR h distributions, it is important not to round up h , which would force 1 to be included in those distributions (this is mathematically incorrect; eq. 7a) when their masses are near 1.

Note that, in the previous simulations, the calculations of M_t (eq. 2c) require values of L_t , random draws of M , and simulated values of mean length at age (eq. 2b). L_t values are derived assuming that random draws of M relate to the most abundant and fully selected age groups (i.e., late juveniles and young adults; Lorenzen 2011). Lorenzen (2011) argues that L_t should be set to an end point, such as the lengths associated with the ages at 50% selection or the ages at 50% maturity, and that M at L_t (and hence at the corresponding age) should be matched to M . In this study, L_t values are lengths associated with the ages at 50% maturity. For each application species (see below), the age at 50% maturity is 1 year according to the references cited in Table 2. Although the age at 50% maturity is 1 year, because of the stochastic nature of growth parameters each combination of L_∞ , K , and t_0 determines a unique growth trajectory. Each growth trajectory is in turn associated with a unique random value of L_t (i.e., length predicted by the VBGF for age-1). Clearly, L_t is a random realization and has a distribution that also contributes to the distribution of natural mortality at age, despite the age at 50% maturity being 1 year.

Note also that the simulated unfished SBPRs involve the reproductive proportions ρ_t , μ_t , and π_m , which are fixed for all replicates, although they are also subject to uncertainty. The reason is that for most application species presented below, the literature does not describe quantitatively how the proportions in question have been variable, nor does it show the age-specific or monthly sample sizes used to calculate the proportions. Otherwise, if large enough samples associated with the calculations of ρ_t , μ_t , and π_m were available, these proportions could be sampled from binomial distributions. Alternatively, the proportions ρ_t , μ_t , and π_m could have been sampled like the growth parameters if studies had shown their seasonal or interannual variations.

Describing priors for the recruitment compensation metrics

Once the simulated priors of κ and h have been obtained, it is necessary to calculate the related summary statistics and estimate parameters of the probability density functions (PDFs) that best characterize those priors. In the present study, the fitdistrplus package in R (Delignette-Muller and Dutang 2015) is used to estimate the parameters that generate the maximum log-likelihood fit to κ and h simulated priors. The choice of candidate PDFs to fit unknown distributions of κ and h is made using the Cullen and Frey plot (also called a skewness-kurtosis graph). The PDF with the greatest loglikelihood as well as the smallest Akaike information criterion and Bayesian information criterion is considered to have the best fit and to provide the best match to an unknown distribution.

¹Supplementary data are available with the article through the journal Web site at <http://nrcresearchpress.com/doi/suppl/10.1139/cjfas-2018-0463>.

Table 2. Application species.

Common name (code, stock ^a)	Scientific name	Min.–max. $T_{\max}^b$	Min.–max. $L_{50\%}^c$	Life-history characterization	Source	
					Growth ^d	Reproduction ^e
Atlantic croaker (ACR, ATL)	<i>Micropogonias undulatus</i>	6–17	17.3–23.3	Medium-bodied and medium-lived	ASMFC 2010a	ASMFC 2010a
Bluefish (BLF, ATL)	<i>Pomatomus saltatrix</i>	7–14	29.87–45.1	Large-bodied and medium-lived	Salerno et al. 2001; Robillard et al. 2009; Northeast Fisheries Science Center 2015	Salerno et al. 2001; Robillard et al. 2008; Northeast Fisheries Science Center 2015
Red snapper (RDS, SEA)	<i>Lutjanus campechanus</i>	45–54	314–335	Large-bodied and long-lived	White and Palmer 2004; SEDAR 41 2017	SEDAR 41 2017
Sheepshead (SPH, SEA)	<i>Archosargus probatocephalus</i>	15–40	236.4–278	Medium-bodied and long-lived	Adams et al. 2018	McDonough et al. 2011
Spot (SPT, ATL)	<i>Leiostomus xanthurus</i>	3–6	186–292	Small-bodied and short-lived	ASMFC 2010b	ASMFC 2010b

^aATL, entire Atlantic Coast; SEA, southeast Atlantic Coast (from Virginia or from North Carolina to East Florida).

^b $T_{\max}$ is longevity (years), with its minimum (min.) and maximum (max.) values observed according to various age-data sources.

^c $L_{50\%}$ is length (cm or mm), with its minimum (min.) and maximum (max.) values observed according to various reproduction-data sources.

^dGrowth includes the von Bertalanffy growth function parameters and the weight–length scales and exponents.

^eReproductive information consists of the age-specific maturity, sex ratios for females, and the estimated proportions of spawning-capable females by month in a year.

Fig. 2. Variations of (A) sex-ratio by age (ρ_t), (B) proportion of mature females by age (μ_t), and (C) proportion of spawning-capable females by month (π_m) for Atlantic croaker (ACR), bluefish (BLF), red snapper (RDS), sheepshead (SPH), and spot (SPT) in the Atlantic Ocean off the US East Coast.

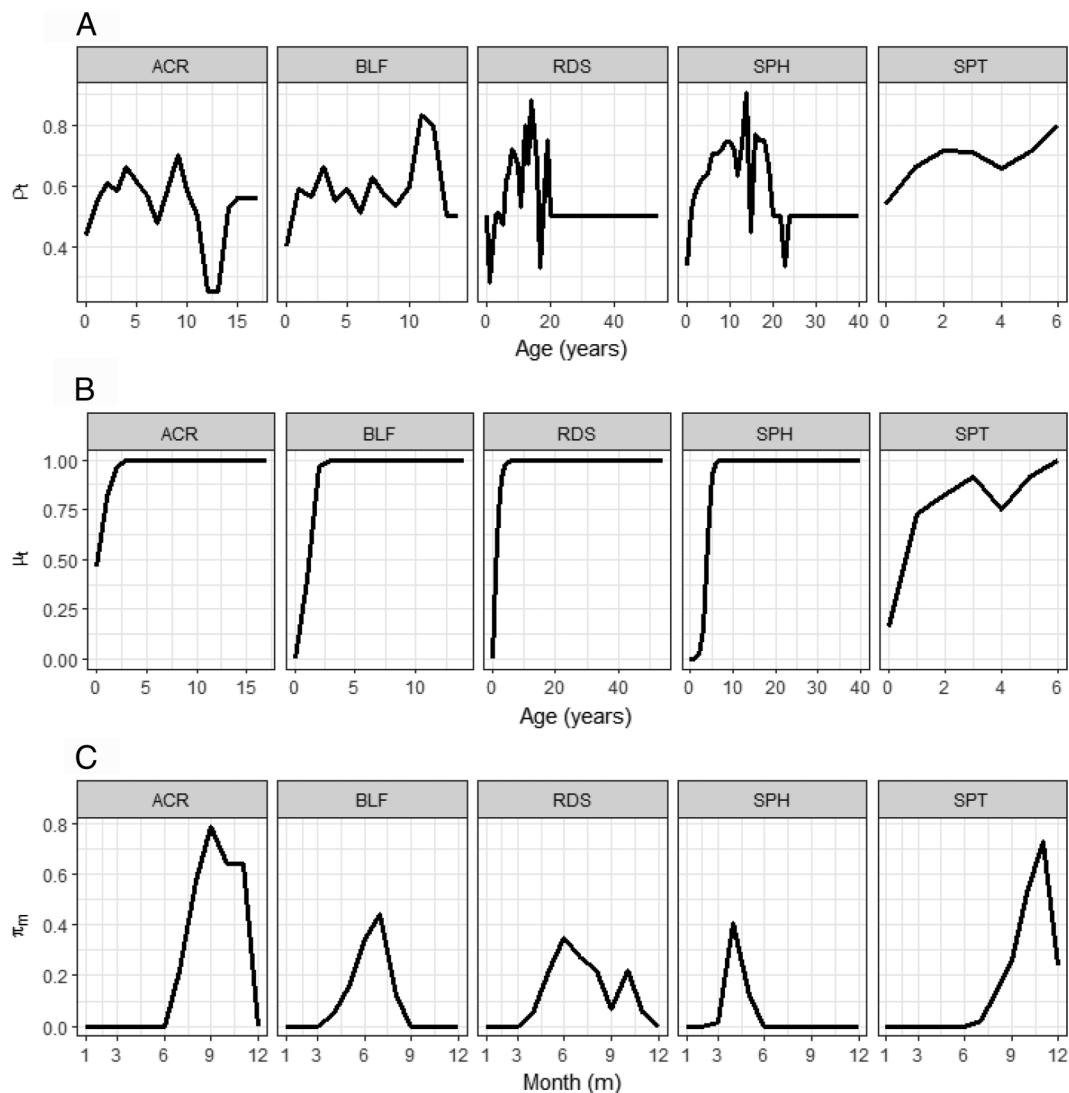
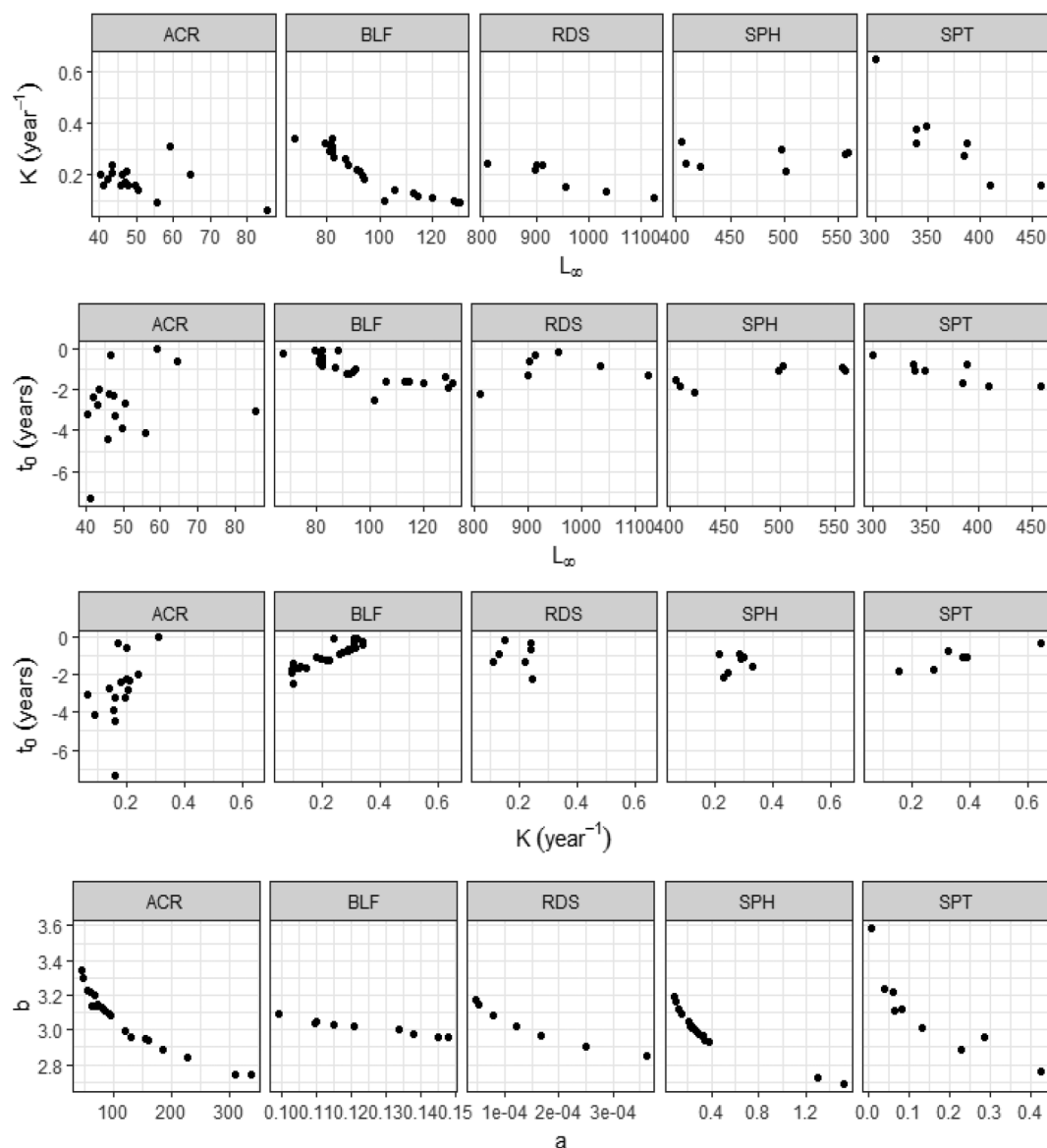


Fig. 3. Relationships between the von Bertalanffy growth function parameters (K versus L_{∞} , t_0 versus L_{∞} , and t_0 versus K) and weight-length coefficients (b versus a) for Atlantic croaker (ACR), bluefish (BLF), red snapper (RDS), sheepshead (SPH), and spot (SPT) in the Atlantic Ocean off the US East Coast (L_{∞} in cm for ACR and BLF, in mm otherwise; to avoid overlap of decimals or scientific notations for a on the x axis of the bottom plots, the original values of a are multiplied by 10 000).



Application of the present method to five finfish species found off the US East Coast

The present method is applied to five finfish species found off the US East Coast (Atlantic croaker, bluefish, red snapper, sheepshead, and spot) using life-history parameters available in the literature, including (peer-reviewed and accepted) stock assessments and life-history reports (Table 2). These species are important resources in their respective fisheries jurisdictions. That is, Atlantic croaker and spot, managed by the Atlantic States Marine Fisheries Commission (ASMFC), are targeted by commercial and recreational fisheries, especially in the mid-Atlantic region, and are important bycatch of shrimp trawl fisheries in the Southeast. The bluefish is one of the most heavily recreationally targeted species in the Atlantic off the US East Coast, managed jointly by the ASMFC and the Mid-Atlantic Fishery Management Council. The red snapper is a commercial and recreational species in the Southeast, where it is federally managed through the South Atlantic Fishery Management Council. The main commercial and recre-

ational fisheries of sheepshead are in the Southeast, especially off the coast of Florida, where the fish fall under Florida's management regulations. On top of their economic and bycatch importance, these species are examined in the present study mainly because their contrasting life histories (Table 2) may help to determine the magnitudes and patterns of simulated recruitment compensation metrics in relation to longevity and asymptotic length.

The parameters used in eqs. 2a–7b apply to the entire East Coast for Atlantic croaker, bluefish, and spot and to the Southeast Atlantic Coast for sheepshead (Virginia to East Florida) and red snapper (North Carolina to East Florida). The reproductive values pertain to females and consist of the age-specific proportions of sex ratio and maturity, as well as the monthly proportions of spawners (Fig. 2). The latter variable is based on the ripe-running stage for Atlantic croaker and spot, the spawning stage for bluefish (Robillard et al. 2008) and red snapper (SEDAR 41 2017), and the spawning-capable stage for sheepshead (McDonough et al.

2011). Whatever the terminologies for these spawning stages, females at those stages are considered spawning capable; spawning biomass pertains to females only.

When growth variables have been found to be sexually dimorphic, only values for females are used; otherwise, values are combined. Figure 3 shows the relationships among the observed growth parameters for the application species. Mean weight at age and unfished SBPRs are calculated in kilograms to facilitate comparison of the recruitment compensation metrics among species.

For each species, uncertainty in growth parameters (Fig. 3), the parameter α (inferred from an empirical regression of eq. 1), and natural mortality M (eq. 2a) are introduced, and Monte Carlo simulations of quantities described by eqs. 2a–7b are performed using the method described previously. Supplementary materials 2¹ gives the Cullen and Frey plot prototypes of the simulated κ and h values. For κ and the R-SRR h of all species, the candidate PDFs are log-normal, gamma, and Weibull distributions. For the BH-SRR h , the candidate PDFs are beta and normal models for Atlantic croaker, bluefish, red snapper, and sheepshead and beta, normal, and uniform for spot.

Sensitivity analyses

The previous simulations employ Pauly's nonlinear estimator of M (eq. 2a), a recruiting age group of 0, and L_r associated with the age at 50% maturity of 1 year. These simulations are herein treated as baseline analyses. Other analyses are conducted to determine how results from baseline analyses are sensitive to the following configurations, denoted SA1 to SA5:

(i) SA1 is similar to the baseline analyses except that eq. 2a is replaced by Hoenig's nonlinear estimator of M (i.e., $M = 4.899T_{\max}^{-0.916}$; Then et al. 2015). Uncertainty in maximum age $T_{\max}$ is characterized by assuming a uniform distribution on the interval from the recorded minimum $T_{\max}$ to the recorded maximum $T_{\max}$ (Table 2).

(ii) SA2 is similar to the baseline analyses, but L_r associated with the age at 50% maturity of 1 year is replaced by values drawn from a uniform distribution on the interval from the minimum estimated length at 50% maturity ($L_{50\%}$) to the maximum estimated $L_{50\%}$ (Table 2). Note that in baseline analyses, L_r and M at the age associated with L_r are conditional on the age at 50% maturity, whereas in SA2, the age at 50% maturity, and hence M at that age, are inferred post hoc given a growth trajectory and knowledge or random realization of L_r .

(iii) SA3 is similar to the baseline analyses except that the age of recruitment used is 1 year instead of 0. In fact, SBPR analyses can be initialized at age-1 ($l_1 = 1$ for age-1; e.g., Martell et al. 2008).

(iv) SA4 is similar to SA1 except that the age of recruitment is 1 year.

(v) SA5 is similar to SA2 except that the age of recruitment is 1 year.

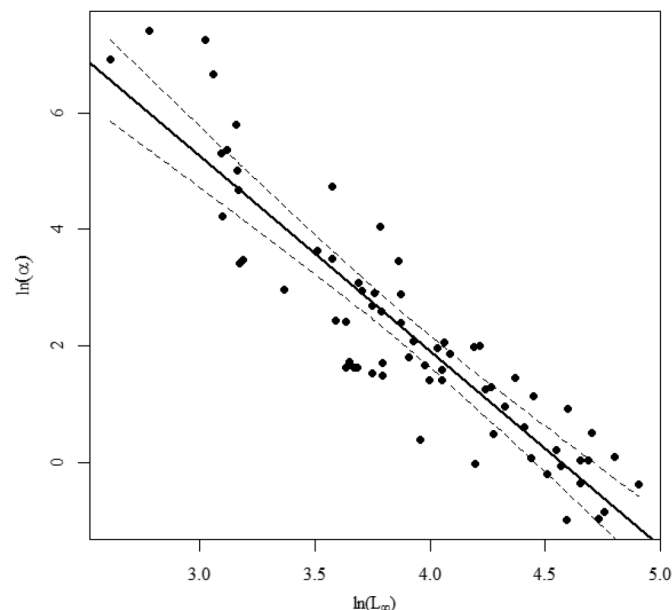
Results

A negative allometry between α and L_∞

The estimated intercept and slope of the log-log-linear relationship between α and L_∞ (Fig. 4, $R^2 = 0.83$, $P < 0.001$) are, respectively, 15.28 (SE = 0.736) and -3.344 (SE = 0.186). The standard deviation of the residuals for the fitted regression eq. 1 is 0.908.

The negative allometry between α and L_∞ leads to an equation relating M to both the growth rate K and α . In fact, rearranging eq. 1 in terms of L_∞ in cm (with $j = 15.28$ and $k = -3.344$) and inserting the new expression into eq. 2a give $M \approx 0.912K^{0.73}\alpha^{0.0987}$. Thus, both α and M relate negatively to L_∞ , but M relates positively to K and α .

Fig. 4. Log-log relationship between the stock-recruit parameter α and asymptotic lengths (L_∞) for various stocks compiled from the literature. The dotted lines indicate the 95% confidence interval of the fitted solid line.



Baseline results

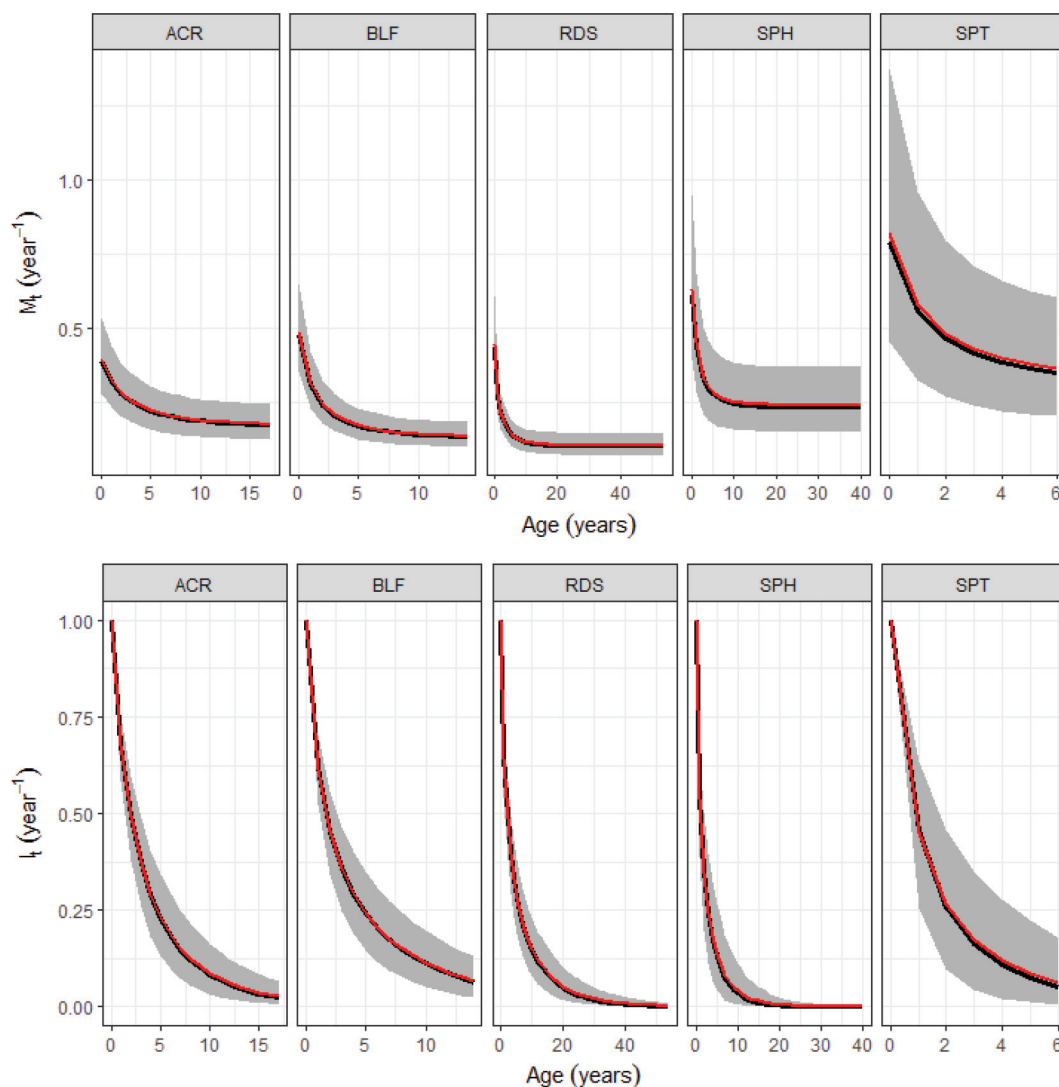
Baseline analyses are simulations using eq. 2a as estimator of M , eq. 2c where reference length L_r is associated with the age at 50% maturity of 1 year, and eqs. 4–5 where the age of recruitment is 0. As expected, larger-bodied and longer-lived species endure the smallest natural mortality rates, resulting in higher survivorship to ages (Fig. 5). For the life-history arguments of the BH-SRR and R-SRR (α , Φ_0 , κ , and h), baseline analyses produce the distributions shown in Figs. 6 and 7 for the application species. Table 3 summarizes the descriptive statistics of these distributions. The distribution of α widens as asymptotic body length decreases, and the statistical moments (mean and variance, hence SD) of α decrease with asymptotic length (Fig. 6A; Table 3). They are greatest for spot, a small-bodied fish, and least for bluefish and red snapper, larger-bodied fishes. The opposite is seen for the unfished SBPR Φ_0 (Fig. 6B; Table 3).

The magnitudes and patterns of κ amalgamate the life-history traits of asymptotic length and longevity (Fig. 6C; Table 3). In general, κ increases with asymptotic length; however, importantly, κ is positively and strongly related to longevity and so is negatively and strongly related to M and α . The magnitudes and patterns of h in relation to asymptotic length and longevity are like the magnitudes and patterns for κ (Fig. 7; Table 3).

The simulated distributions of κ are right-skewed for all species (Fig. 6C). Of the candidate PDFs fitted to these distributions, the log-normal PDF matches them best because its maximum log-likelihood is highest and its information criteria are lowest (Supplementary Fig. S2.5 and Table S2.1). The “meanlog” (μ) and “sdlog” (σ) parameters that generate the maximum log-likelihood fit of the log-normal PDF to κ distributions are, respectively, 2.2 and 0.94 for Atlantic croaker, 2 and 0.9 for bluefish, 2.72 and 0.96 for red snapper, 2.49 and 1.08 for sheepshead, and 1.69 and 0.95 for spot.

The simulated distributions of the BH-SRR steepness are left-skewed for Atlantic croaker, bluefish, red snapper, and sheepshead; the BH-SRR steepness of spot is equally important between 0.2 and 0.98 (Fig. 7, left panel). For the first four species, the beta PDF matches their steepness distributions better than does the normal PDF (Supplementary Fig. S2.6 and Table S2.2¹). The shape

Fig. 5. Variations of median (black line), mean (red line), and 95% uncertainty envelope (gray area) of natural mortality (M_t) by age and survivorship to an age (l_t) obtained with baseline analyses (employing eq. 2a, age-0 as age of recruitment, and L_r corresponding to the age at 50% maturity of 1 year) for Atlantic croaker (ACR), bluefish (BLF), red snapper (RDS), sheepshead (SPH), and spot (SPT) in the Atlantic Ocean off the US East Coast. [Colour online.]



parameters of the beta PDFs, shape1 and shape2, are, respectively, 4.13 and 2.01 for Atlantic croaker, 3.88 and 2.31 for bluefish, 5.23 and 1.71 for red snapper, and 3.64 and 1.51 for sheepshead. For spot, either the symmetric beta PDF (shape1 = 3.04 and shape2 = 2.37) or the uniform PDF is more suitable than the normal PDF for fitting the BH-SRR steepness data.

Finally, the simulated distributions of the R-SRR steepness are, as for κ , right-skewed for all species (Fig. 7, right panel). Compared with other candidate PDFs tested, the log-normal PDF best matches those R-SRR steepness distributions (Supplementary Fig. S2.6 and Table S2.3¹). The “meanlog” (μ) and “sdlog” (σ) parameters that generate the maximum log-likelihood fit of the log-normal PDF to the R-SRR steepness distributions are, respectively, 0.14 and 0.74 for Atlantic croaker, -0.009 and 0.72 for bluefish, 0.57 and 0.77 for red snapper, 0.38 and 0.86 for sheepshead, and -0.26 and 0.76 for spot.

There exist equations expressing the means, medians, modes, SD, skewness, and kurtosis of κ and h in Table 3 as functions of the PDF parameters shown in Supplementary Tables S2.1–S2.3¹ (Law and Kelton 1982; Forbes et al. 2011).

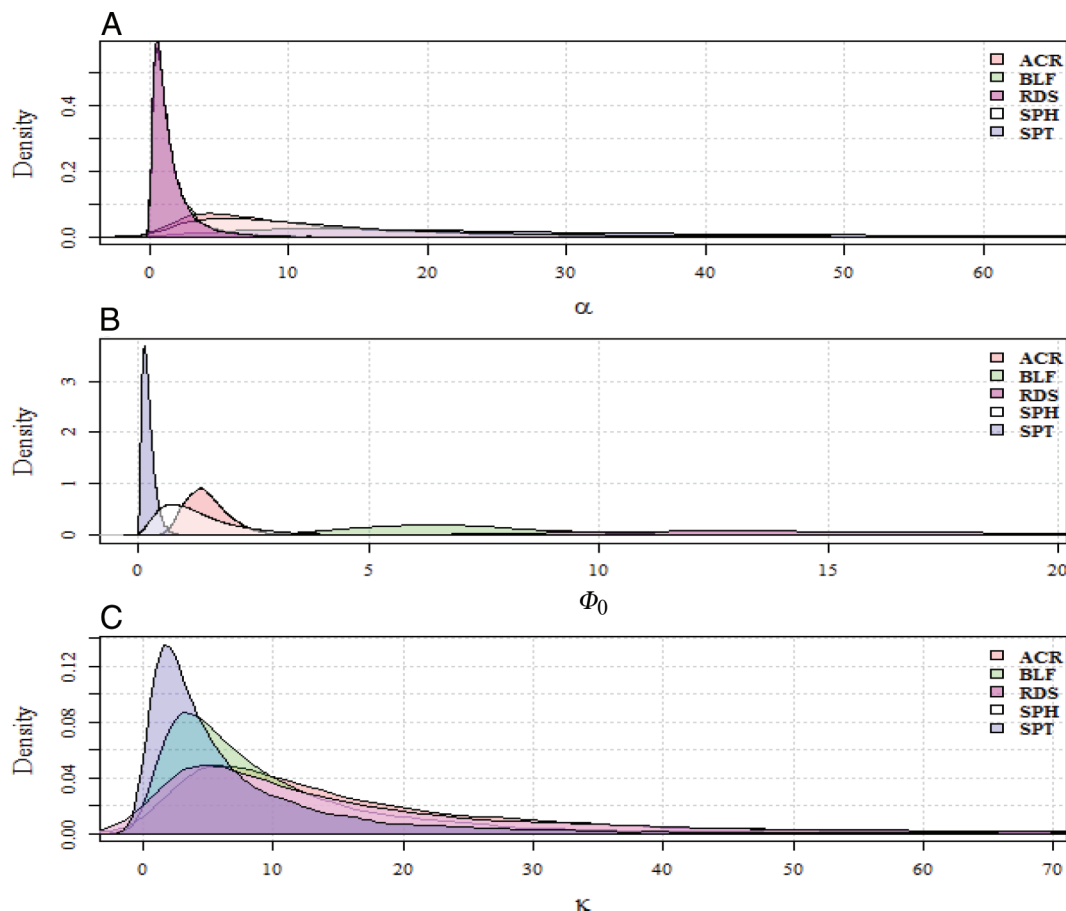
Sensitivity results

To ease interpretation of sensitivity results, consider that, for each species, the distributions of growth parameters vary little within and among simulation types (i.e., baseline versus each sensitivity analysis and sensitivity analyses versus each other). Likewise, α distributions inferred therefrom vary little.

The sensitivities of baseline results to the choices of Hoenig’s nonlinear estimator of M , reference lengths (within the range of lengths at 50% maturity) for calculating M at age (eq. 2c), and age-1 as age of recruitment are shown in Fig. 8 and Table 4. Figure 8 illustrates the simulated distributions of steepness (those of Φ_0 and κ are not shown to preserve space, but they are available upon request); Table 4 gives some descriptive statistics of κ and h . Sensitivity results also indicate that the beta PDF and log-normal PDF best match, respectively, the BH-SRR h and R-SRR h . For the BH-SRR h of spot, however, a uniform PDF is also suitable.

When Hoenig’s nonlinear estimator of M is used (SA1), the distributions of h for the BH-SRR and R-SRR of Atlantic croaker, bluefish, and spot become more and more right-skewed, and the means and medians of κ and h are less than the same values from

Fig. 6. Simulated density distributions of (A) the slope at the origin of the Beverton–Holt stock–recruit relationship (BH-SRR) and Ricker stock–recruit relationship (R-SRR) curves (α) inferred from the relationship illustrated in Fig. 4, (B) the unfished spawning-stock biomass per recruit (Φ_0), and (C) the recruitment compensation ratio (κ) for Atlantic croaker (ACR), bluefish (BLF), red snapper (RDS), sheepshead (SPH), and spot (SPT) in the Atlantic Ocean off the US East Coast. For plots (A) and (C), the scales of the x axis are 10 times reduced, by cutting the right tails off, to increase the resolutions of the plots. The densities of α for BLF and RDS and of κ for SPH and RDS are virtually similar and superimposed, but the densities of Φ_0 for all species are distinct. The values of Φ_0 and κ are based on baseline analyses (employing eq. 2a, age-0 as age of recruitment, and L_r corresponding to the age at 50% maturity of 1 year). [Colour online.]



the baseline analyses. The opposite is observed for red snapper and sheepshead.

Except for Atlantic croaker, use of L_r values randomly drawn from uniform distributions on the range of lengths at 50% maturity (SA2) shifts the distributions of κ and h to the left of these parameter ranges, producing smaller values of κ and h . For Atlantic croaker, this means that L_r values in SA2 are smaller than L_r values employed in the baseline simulations. Small L_r values induce smaller values of M_t and improve survivorships to older ages at which individuals become heavier; this leads to higher values of Φ_0 , κ , and h . In short, smaller L_r values induce the increase in the compensatory capacity of a stock. Other species experience the opposite (higher L_r values in SA2 than in baselines, increased M_t , decreased survivorships and compensatory capacities).

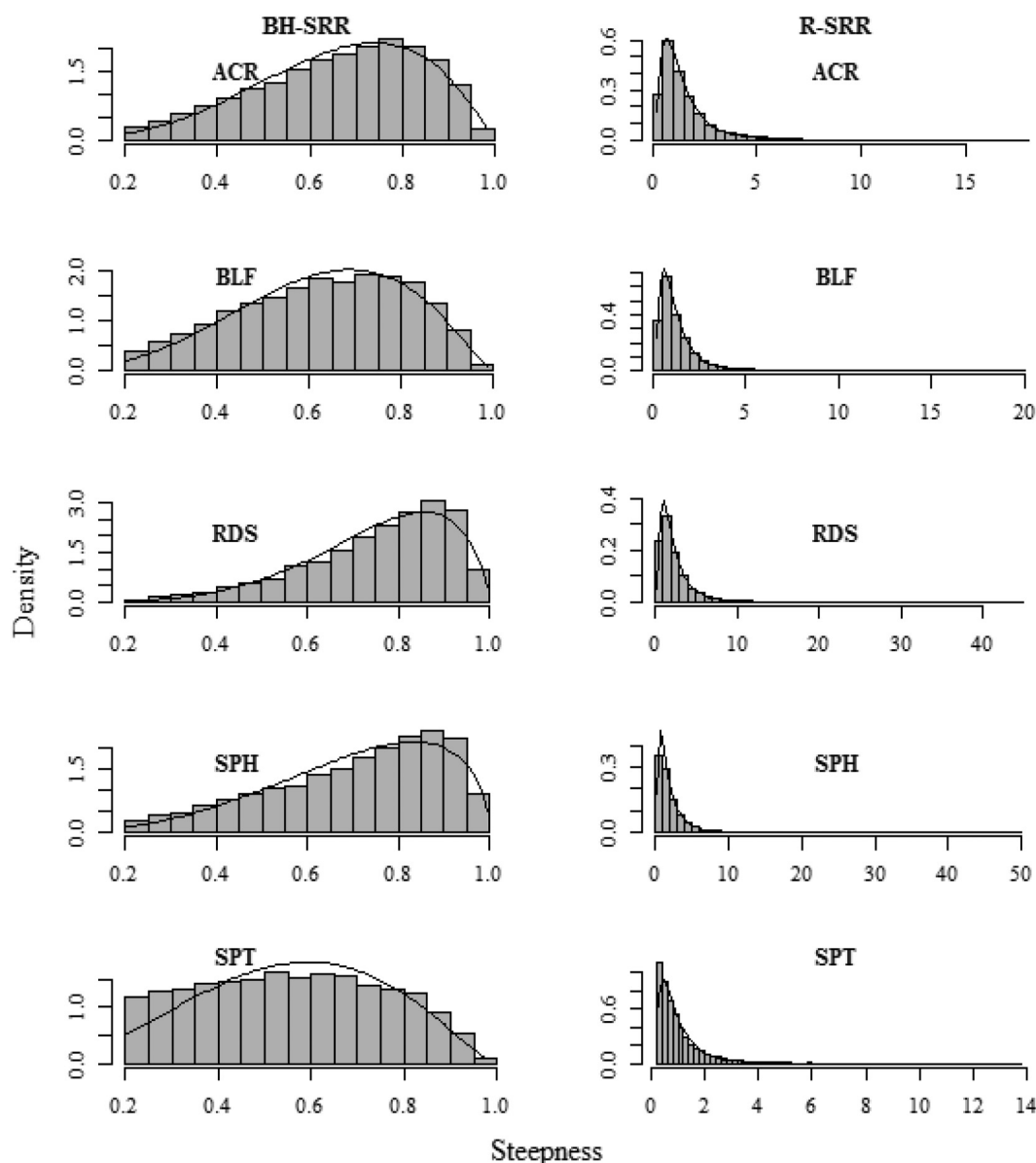
When the age of recruitment of 0 is replaced by age-1 (SA3), age-1 survivorship is maximum (1) whereas, in baseline simulations, it represents a fraction of the survivorship of 1 at age-0. Age-1 survivorship in SA3 results in higher survivorship to successive ages. Higher survivorships in turn increase Φ_0 , κ , and h . The same effect of using age-1 as the age of recruitment is perceptible for sensitivity analyses SA4 and SA5, but can be dampened by joint use of Hoenig's nonlinear equation for M (SA4) and of L_r values randomly drawn from uniform distributions on the range of lengths at 50% maturity (SA5).

Discussion

Although a negative allometry between α and L_∞ has been demonstrated (Denney et al. 2002; Goodwin et al. 2006; Hall et al. 2006; Munyandorero 2018; but see Foss-Grant et al. 2016 for an exception), it has been generally overlooked or ignored for estimating α , κ , and h for a species. Yet for stock assessments and simulations (e.g., Piner et al. 2011; Wiedenmann et al. 2013; Wetzel and Punt 2015; Dichmont et al. 2016), such an allometry may prove as useful as the popularly used estimators of M (e.g., Then et al. 2015). M estimators have helped in estimating M values consistent with species' life histories, departing from a practice used in the 1970s–1990s, whereby M was arbitrarily set to 0.2 for almost all exploited populations despite differences in their life-history traits (e.g., Goodwin et al. 2006). Likewise, in fishery models, the development and use of species-specific values of α , κ , or h following the above-mentioned allometry would overcome a contemporary practice of employing α , κ , or h values obtained for a group of species whose life histories are believed to be similar even though they are different.

The priors of κ and h based on the method presented here must be applied only if analyses rely on the unfished SBPR in kilograms and stock assessments or simulations where, in SRRs, the spawning-stock biomass is in metric tonnes and recruitment is in

Fig. 7. Plots of the beta and log-normal probability density functions (solid lines) fitted to the simulated steepness values (histograms) of the Beverton–Holt stock–recruit relationship (BH-SRR; left panel) and Ricker stock–recruit relationship (R-SRR; right panel) obtained with baseline analyses (employing eq. 2a, age-0 as age of recruitment, and L_r corresponding to the age at 50% maturity of 1 year) for Atlantic croaker (ACR), bluefish (BLF), red snapper (RDS), sheepshead (SPH), and spot (SPT) in the Atlantic Ocean off the US East Coast.



thousands of individuals. The reason is that the units of the parameter α (recruits per kilogram of spawning-stock biomass, recruits per metric tonne of spawning-stock biomass, recruits per thousand metric tonnes of spawning-stock biomass, and recruits per number of eggs), the unfished SBPR (kg or g), and the unfished total number of eggs per recruit greatly influence the locations of the distribution mass of κ and h (personal observations). Thus, in principle, if analysis preferences are given to age-structured models where recruitment is the number of individuals and the parental stock is total number of eggs, then κ and h priors must be obtained using the unfished number of eggs per recruit and eq. 1 (to be developed), where the unit of α must be recruits per number of eggs. Likewise, priors of κ and h for analyses employing stage-structured models must rely on appropriate units of α to develop the regression of eq. 1 and on appropriate unfished composite spawning stock per recruit (Munyanadorero 2015).

This study is in line with the previous studies by Hall et al. (2006), Rochet et al. (2011), Thorpe et al. (2015), and Munyanadorero (2018) in using an α - L_∞ relationship to infer α and by Munyanadorero (2018) in using an α - L_∞ relationship for developing priors of κ and h . Because the proposed method integrates a meta-analysis of an α - L_∞ relationship (gained from other stocks irrespective of life-history similarity) and life-history information of a species to generate the related κ and h priors, it can be considered a hybrid of classical meta-analytic approaches (Myers et al. 1999, 2002; Shertzer and Conn 2012; Punt and Dorn 2014; Foss-Grant et al. 2016; Thorson et al. 2019) and reproductive ecology methods (Mangel et al. 2010; Simon et al. 2012; Brodziak et al. 2015; Cortés 2016; Kai and Fujinami 2018).

Common to the method developed herein and classical meta-analyses is that both methods rely on a proxy of compensation ratio as defined in Walters and Martell (2004) and van Poorten

Table 3. Descriptive statistics from a simulation run for the inferred slopes at the origin of stock–recruit curves α (recruits per thousand metric tonnes of spawning-stock biomass), the unfished spawning biomass per recruit Φ_0 (kg), the compensation ratio κ (unitless), and the Beverton–Holt stock–recruit relationship (BH-SRR) and Ricker stock–recruit relationship (R-SRR) steepness h (unitless) for the species investigated.

Statistic	Atlantic croaker	Bluefish	Red snapper	Sheepshead	Spot
α					
Mean (SD)	13.08 (15.09)	1.63 (1.77)	1.59 (1.74)	15.72 (16.93)	36.91 (39.54)
Median	8.70	1.08	1.07	10.47	24.43
Mode	0.39	0.04	0.04	0.44	0.70
Skewness	5.29	3.78	4.04	3.95	3.97
Kurtosis	62.52	28.95	32.88	34.74	31.72
Min.–max.	0.39–340.89	0.04–25.75	0.04–29.72	0.44–327.96	0.70–727.47
90% probable range	1.95–38.37	0.24–4.83	0.24–4.65	2.41–46.41	5.51–110
Φ_0					
Mean (SD)	1.06 (0.36)	6.78 (2.04)	15.10 (5.41)	1.36 (0.98)	0.21 (0.14)
Median	1.01	6.60	14.30	1.13	0.18
Mode	0.25	1.63	3.12	0.02	0.0014
Skewness	0.79	0.53	0.91	1.95	1.45
Kurtosis	4.00	3.31	4.31	10.07	6.38
Min.–max.	0.25–3.64	1.63–16.12	3.12–48.61	0.02–12.33	0.0014–1.32
90% probable range	0.57–1.71	3.77–10.46	7.74–25.08	0.30–3.24	0.05–0.49
κ					
Mean (SD)	13.98 (17.77)	11.27 (12.83)	24.21 (29.21)	21.66 (30.99)	8.87 (12.10)
Median	8.84	7.18	15.18	11.96	5.07
Mode	1.006	1.0003	1.02	1.005	1.002
Skewness	6.25	3.89	4.42	5.71	5.44
Kurtosis	82.04	29.47	39.52	70.52	56.83
Min.–max.	1.006–391.20	1.0003–194.95	1.02–511.75	1.005–770.69	1.002–245.27
90% probable range	1.97–41.89	1.71–34.61	3.09–75.4	2.003–71.96	1.30–28.23
BH-SRR h					
Mean (SD)	0.66 (0.18)	0.63 (0.183)	0.76 (0.16)	0.71 (0.19)	0.56 (0.20)
Median	0.69	0.64	0.79	0.75	0.56
Mode ^a	0.85	0.76	0.89	0.91	0.57
Skewness	–0.48	–0.30	–0.94	–0.71	0.03
Kurtosis	2.41	2.22	3.40	2.62	1.95
Min.–max. ^b	0.20–0.99	0.20–0.98	0.20–0.99	0.20–0.99	0.20–0.98
90% probable range	0.33–0.91	0.30–0.90	0.44–0.95	0.33–0.95	0.24–0.88
R-SRR h					
Mean (SD)	1.53 (1.39)	1.30 (1.01)	2.38 (2.12)	2.13 (2.21)	1.05 (1.04)
Median	1.14	0.97	1.76	1.46	0.73
Mode	0.83	0.62	1.04	0.53	0.28
Skewness	3.89	2.76	3.02	3.64	3.65
Kurtosis	34.55	15.99	19.59	29.06	27.19
Min.–max. ^b	0.20–23.71	0.20–13.58	0.20–29.4	0.20–40.79	0.20–16.32
90% probable range	0.34–3.97	0.31–3.41	0.49–6.35	0.35–6.12	0.24–2.95

^aFor the BH-SRR, Shertzer and Conn (2012) argue that if data are informative, “a prior distribution of steepness informs the estimation process because best estimate occurs at the mode”.

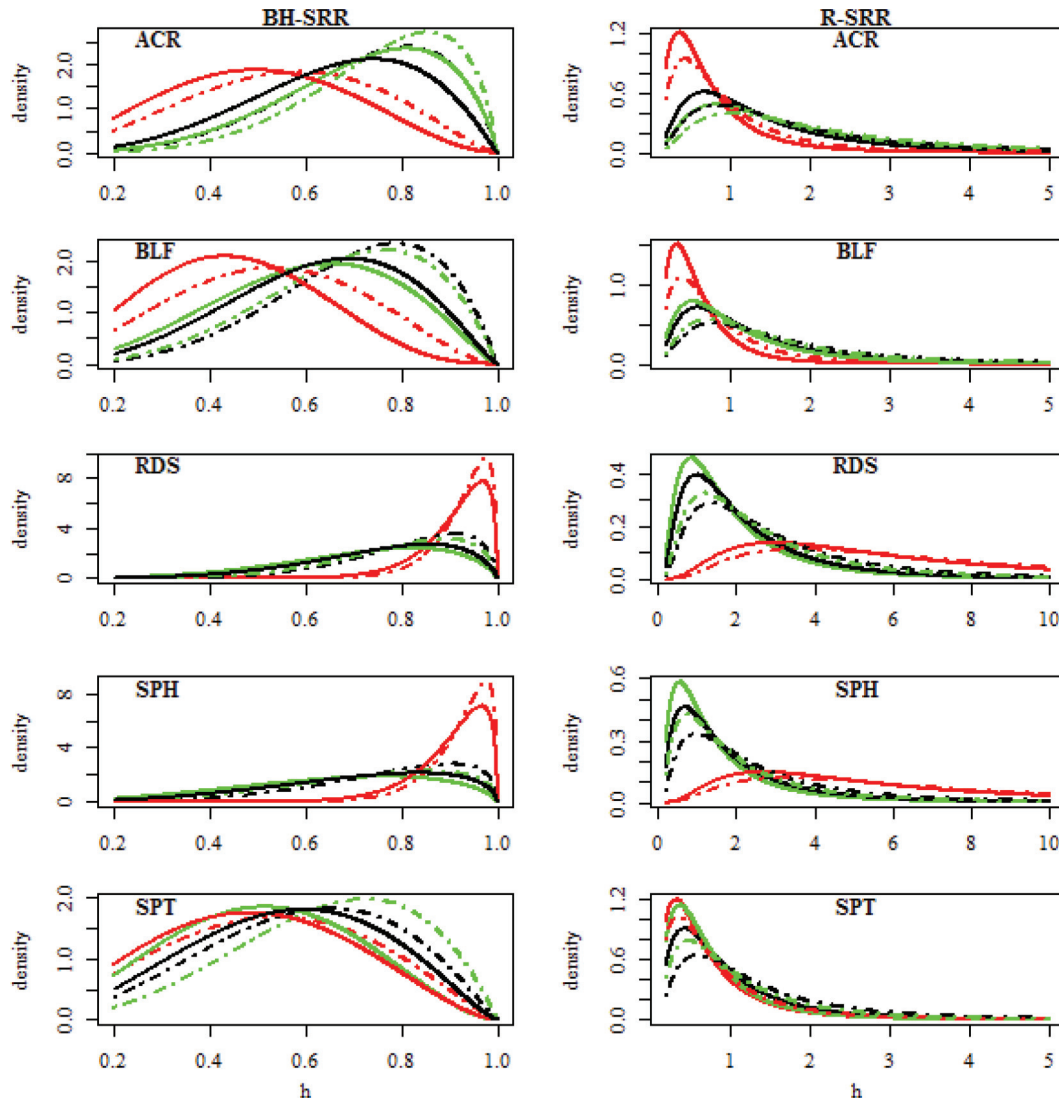
^bThe interval min.–max. is open at the lower bound.

et al. (2018). In fact, the maximum rate of recruit production is a proxy of egg-to-recruit survival for the low number of total eggs produced (i.e., the number of total eggs produced approaches zero), and the reciprocal of the unfished SBPR ($1/\Phi_0$) is a proxy of egg-to-recruit survival for an unfished state. But the recruitment compensation metrics generated using classical meta-analyses are fixed for a group of species believed to have similar life histories. In contrast, the method developed in this study holds constant the above-mentioned α – L_∞ relationship, but the recruitment compensation metrics are species-specific owing to species-specific unfished SBPR and inferred parameter α . Reproductive ecology methods, on the other hand, aim to fulfil the input requirement (egg-to-recruit survival for the near-zero biomass of spawners) of Walters and Martell’s (2004) and van Poorten et al.’s (2018) definition of the compensation ratio of a species. This goal is apparently attainable for sharks (Cortés 2016; Kai and Fujinami 2018). For finfish species, the reproductive ecology methods have so far inferred egg-to-recruit mortalities using meta-analytic relationships

between early-stage natural mortality and dry weight (Mangel et al. 2010; Simon et al. 2012; Brodziak et al. 2015), because such mortality estimates were unavailable or deemed inconsistent. Therefore, in addition to the unfished SBPR, the method developed and the existing reproductive ecology applications to single-species stocks use meta-analyses alike, the only difference being that those meta-analyses relate, respectively, to recruited age-groups and prerecruit stages.

The method presented in this study propagates uncertainty in α , M , Φ_0 , κ , and h , considering that a species has exhibited many growth trajectories as reflected in a variety of VBGF parameters and weight–length scales and exponents. But all trios (L_∞ , K , t_0) and all pairs (a , b) are unique and valid, together characterizing scientific uncertainty. Such parameters may not always be available. It is, however, still possible to perturb the calculations of α , M , Φ_0 , κ , and h . For example, when an age-structured stock assessment is initiated, at least one VBGF and one weight–length relationship are fitted. In that case, the calculations employing eqs. 2a–7b can

Fig. 8. Plots of the beta and log-normal probability density functions (PDFs) fitted to the simulated steepness values of the Beverton–Holt stock–recruit relationship (BH-SRR; left panel) and Ricker stock–recruit relationship (R-SRR; right panel) for showing the sensitivities of PDFs from baseline analyses (solid black line) to various configurations of simulations (denoted SA1 to SA5 in the text) for Atlantic croaker (ACR), bluefish (BLF), red snapper (RDS), sheepshead (SPH), and spot (SPT) in the Atlantic Ocean off the US East Coast. SA1 (solid red line), Hoenig nonlinear estimator of M replaces eq. 2a in baseline; SA2 (solid green line), values of reference length L_r are drawn from a uniform distribution on the interval from the minimum estimated length at 50% maturity to the maximum estimated length at 50% maturity; SA3 (broken-dotted black line), age of recruitment of 0 (in baseline) is replaced by age-1; SA4 (broken-dotted red line), similar to SA1 except that the age of recruitment is 1 year; and SA5 (broken-dotted green line), similar to SA2 except that the age of recruitment is 1 year. [Colour online.]



be perturbed by generating relevant statistics of variability in growth parameters.

Along with the α - L_∞ relationship, a key step to deriving κ and h is the calculation of the unfished SBPR (eqs. 4 and 5). A calculation methodology in this respect integrates the statistic λ_t (eq. 2e) to account for the prolonged, even year-round, spawning activity that many fish populations exhibit. Munyandorero (2018) described and discussed extensively the statistic λ_r . This statistic accommodates various configurations of the calculation of the SBPR depending on the annual reproductive dynamics of the focal species and allows the generalization of Gabriel et al.'s (1989) and Haddon's (2001) equations for calculating the SBPR.

This study shows that the compensatory (density-dependent) recruitment is strongest in long-lived species, a property previously noted by many researchers (e.g., Myers et al. 2002; Goodwin et al. 2006; Andersen and Beyer 2015; Cortés 2016) and adopted by

Coggins et al. (2007), Allen et al. (2013), and Gwinn et al. (2015). In other words, κ and h are positively related to life span and hence negatively related to natural mortality and maximum rate of recruit production. But many other studies theorized, assumed, or found the opposite (He et al. 2006; Garrison et al. 2011; Mangel et al. 2010, 2013; Wiedenmann et al. 2013; Wetzel and Punt 2015, 2016) or indicated that steepness was independent of life-history traits of longevity (Shertzer and Conn 2012). Do these results and assumptions suggest that recruitment compensation is highest in long-lived species, highest in short-lived species, or similar across life histories? This is an open question that warrants further investigation.

Compared to baseline analysis results, the sensitivity analysis using Hoenig's nonlinear estimator of M shifts the distribution of h to the left of the h range for short-lived and medium-lived species and to the right for long-lived species. The reason for these

Table 4. Selected descriptive statistics from a simulation run of sensitivity analyses (SA1–SA5) for the compensation ratio κ (unitless) and the Beverton–Holt stock–recruit relationship (BH-SRR) and Ricker stock–recruit relationship (R-SRR) steepness h (unitless) for Atlantic croaker (ACR), bluefish (BLF), red snapper (RDS), sheepshead (SPH), and spot (SPT) in the Atlantic Ocean off the US East Coast.

Sensitivity	Statistic	ACR	BLF	RDS	SPH	SPT
κ						
SA1	Mean (SD)	6.22 (8.47)	4.63 (5.73)	89.81 (104.04)	83.41 (103.07)	6.65 (10.8)
	Median	3.51	2.79	58.27	52.47	3.16
	Min.–max.	1.001–137.43	1.001–91.54	1.41–2088.01	1.87–1998.73	1.001–173.65
SA2	Mean (SD)	18.82 (22.34)	10.1 (11.82)	20.26 (25.32)	16.30 (26.38)	6.65 (9.26)
	Median	12.14	6.34	12.47	8.72	3.75
	Min.–max.	1.005–421.19	1.001–193.59	1.003–563.97	1.005–902.12	1.001–210.01
SA3	Mean (SD)	18.71 (21.85)	17.13 (20.95)	36.23 (46.904)	31.91 (45.77)	13.37 (17.39)
	Median	12.09	10.90	22.79	17.74	7.98
	Min.–max.	1.003–313.49	1.003–353.21	1.041–1210.80	1.01–817.42	1.001–398.24
SA4	Mean (SD)	8.42 (11.3)	6.96 (9.15)	111.89 (124.93)	107.68 (132.03)	8.03 (13.09)
	Median	4.85	4.05	73.32	68.19	3.94
	Min.–max.	1.001–188.22	1.001–166.11	1.39–1960.65	1.051–3904.95	1.001–283.96
SA5	Mean (SD)	24.19 (30.19)	15.33 (19.07)	30.95 (38.92)	24.32 (38.45)	10.52 (13.96)
	Median	15.41	9.83	19.45	13.19	6.10
	Min.–max.	1.002–691.12	1.002–626.48	1.023–851.27	1.001–1323.49	1.001–271.73
BH-SRR h						
SA1	Mean (SD)	0.49 (0.19)	0.44 (0.18)	0.91 (0.08)	0.90 (0.08)	0.48 (0.20)
	Median	0.47	0.41	0.94	0.93	0.44
	Min.–max.	0.2001–0.972	0.2001–0.958	0.26–0.998	0.32–0.998	0.2001–0.978
SA2	Mean (SD)	0.72 (0.17)	0.60 (0.19)	0.72 (0.17)	0.65 (0.20)	0.50 (0.19)
	Median	0.75	0.61	0.76	0.69	0.48
	Min.–max.	0.2008–0.991	0.2002–0.98	0.2005–0.993	0.2008–0.996	0.2001–0.981
SA3	Mean (SD)	0.72 (0.17)	0.70 (0.17)	0.81 (0.14)	0.77 (0.17)	0.64 (0.19)
	Median	0.75	0.73	0.85	0.82	0.67
	Min.–max.	0.2005–0.987	0.2004–0.989	0.207–0.997	0.2001–0.995	0.2002–0.99
SA4	Mean (SD)	0.55 (0.20)	0.52 (0.19)	0.93 (0.07)	0.92 (0.074)	0.52 (0.21)
	Median	0.55	0.50	0.95	0.94	0.50
	Min.–max.	0.2001–0.98	0.2001–0.977	0.26–0.998	0.208–0.999	0.2001–0.99
SA5	Mean (SD)	0.76 (0.16)	0.68 (0.18)	0.79 (0.14)	0.72 (0.19)	0.59 (0.20)
	Median	0.79	0.71	0.83	0.77	0.60
	Min.–max.	0.2004–0.994	0.2002–0.994	0.204–0.995	0.2001–0.997	0.2002–0.986
R-SRR h						
SA1	Mean (SD)	0.80 (0.78)	0.64 (0.56)	6.82 (5.85)	6.40 (5.76)	0.82 (0.94)
	Median	0.55	0.45	5.17	4.75	0.503
	Min.–max.	0.2001–10.27	0.2001–7.42	0.26–90.53	0.33–87.42	0.2001–12.383
SA2	Mean (SD)	1.95 (1.71)	1.18 (1.03)	2.06 (1.88)	1.69 (1.89)	0.84 (0.83)
	Median	1.47	0.88	1.51	1.13	0.58
	Min.–max.	0.2008–25.15	0.2002–13.51	0.2005–31.77	0.2008–46.26	0.2001–14.42
SA3	Mean (SD)	1.95 (1.69)	1.81 (1.61)	3.28 (3.04)	2.91 (3.02)	1.47 (1.39)
	Median	1.47	1.35	2.44	2.00	1.05
	Min.–max.	0.2005–19.86	0.2004–21.85	0.21–58.54	0.2001–42.76	0.2002–24.05
SA4	Mean (SD)	1.01 (0.98)	0.87 (0.83)	8.15 (6.80)	7.85 (6.99)	0.96 (1.08)
	Median	0.71	0.61	6.21	5.86	0.60
	Min.–max.	0.2001–13.21	0.2001–11.95	0.26–86.09	0.208–149.39	0.2001–18.35
SA5	Mean (SD)	2.40 (2.14)	1.65 (1.47)	2.89 (2.63)	2.33 (2.57)	1.21 (1.17)
	Median	1.78	1.24	2.15	1.57	0.85
	Min.–max.	0.2004–37.38	0.2002–34.56	0.204–44.17	0.2001–62.86	0.2002–17.71

Note: SA1: Hoenig nonlinear estimator of M replaces eq. 3a in baseline analyses; SA2: values of reference length L_r are drawn from a uniform distribution on the interval from the minimum estimated length at 50% maturity to the maximum estimated length at 50% maturity; SA3: age of recruitment of 0 (in baseline) is replaced by age-1; SA4: similar to SA1 except that the age of recruitment is 1 year; and SA5: like SA2 except that the age of recruitment is 1 year.

patterns is that, in comparison with Pauly's nonlinear equation (eq. 2a), Hoenig's nonlinear equation inflates M for short- to medium-lived species (expectedly experiencing higher natural mortality) and yields lower M estimates for long-lived species, which expectedly endure lower levels of natural mortality (Then et al. 2015). Thus, the increased levels of M reduce Φ_0 , κ , and h for Atlantic croaker, bluefish, and spot. For red snapper and sheepshead, the reduced levels of M increase Φ_0 , κ , and h .

The proposed method has four important caveats. First, it relies on an α – L_∞ relationship assumed to be stationary and known. Thus, the underlying estimates of stock–recruit model outputs of

spawning-stock biomass and recruitment as well as L_∞ are implicitly considered to represent the state of nature. Yet the stock–recruit time series rarely persist across analyses. Should new α and L_∞ pairs be added or different ones be used, then the α – L_∞ relationship would change; Φ_0 , κ , and h would change accordingly.

The second limitation of this method is the implication of the negative relationship between α and L_∞ . Short-lived species with large L_∞ values have low estimates of α (even though, in theory, their α values are high) and vice versa for long-lived species with small L_∞ values. Therefore, the method is useful if a species is

short-lived and small-bodied, medium-lived and medium-bodied, or long-lived and large-bodied. It should be used judiciously for medium-lived and large-bodied species, long-lived and small-bodied species, and short-lived and large-bodied species. But the limitation implied by an α - L_{∞} relationship can be circumvented by using a negative α - $T_{\max}$ relationship, provided the values of α and $T_{\max}$ are stock specific.

Third, this study does not incorporate uncertainty in reproductive proportions (sex ratio and proportions of mature females by age, as well as monthly proportions of spawning-capable females), in large part because the variability in these proportions is limited for the application species. Yet, interannual and seasonal differences in such variables are obvious in marine populations (e.g., Lowerre-Barbieri et al. 2011). Future articles addressing variability in reproductive parameters of the application species are warranted and would allow better characterization of κ and h priors for those species.

Finally, baseline and sensitivity results show that κ and h priors are sensitive, separately or jointly, to an M estimator, age of recruitment, and lengths at 50% maturity (L_r). Those priors are also sensitive to other nonreported aspects, including (i) Monte Carlo configurations (e.g., deterministic M and α versus their sampling from log-normal distribution errors) and (ii) outliers among L_{∞} values. Regarding the latter, L_{∞} outliers shift the priors of κ and h to the left for smaller L_{∞} estimates and to the right for larger L_{∞} estimates. For stock assessment perspectives, the observed sensitivity issues call for transparency about and relevance of input parameters used to develop priors of κ and h .

All in all, the priors of κ and h obtained for the application species using the method developed herein can inform these species' fishery models. The method in question requires fewer input parameters than the reproductive ecology method while also producing recruitment compensation metrics for single species. Unlike prerecruit growth and survival rates needed for reproductive ecology analyses, the input requirements of the proposed method (an allometry between α and L_{∞} as well as growth and reproductive variables of the recruited age groups) can be easily documented for most populations, rendering this method useful. Essentially, the present method is a snapshot of the reproductive ecology method (in that the egg-to-recruit survival for the very low spawning-stock size is approximated by the parameter α inferred from the α - L_{∞} relationship), but undoubtedly is an improvement over classical meta-analyses of recruitment compensation metrics. For this reason, the statistics of κ and h generated by this method should be used to inform stock assessments and simulations instead of the so-called best (or reasonable) guesses of κ and h , or κ and h values generated from classical meta-analyses, because these latter analyses treat species alike even though they are fundamentally different.

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Appendix A. Sampling procedures and sampling distribution errors

Random draws in this study deal with the weight-length scales (a) and exponents (b), VBGF parameters (L_∞ , K , and t_0), maximum rate of recruit production (α), and natural mortality (M). Whether these parameters are observed data or sampled data, they hereinafter keep the previous notations. However, in sampling pseudocodes, these parameters have different object names.

Consider a matrix **W** that has d rows (number of observations) and two columns, the first column for a and the second one for b . Consider also a matrix **G** that has g rows (number of observations) and three columns for, respectively, L_∞ , K , and t_0 . In the following pseudocodes, **W** and **G** denote the object names of the matrices **W** and **G**.

Uncertainty in a , b , L_∞ , K , and t_0 is incorporated through two sampling steps using R version 3.4.3 (R Core Team 2018). The first step is the joint resampling, with replacement, of a and b on the one hand and L_∞ , K , and t_0 on the other. This step does not assume probability distribution errors; it is necessary and may be enough for characterizing uncertainty in growth parameters. The following pseudocodes are used:

```
# for a and b
a.s = c(); b.s = c()

for (n in 1: r) # r is the number of replicates, e.g., r = 10 000
{
  boot.W = sample(d, size = d, replace = T)

  a.s [n] = mean(W[boot.W,1]) # a
  b.s [n] = mean(W[boot.W,2]) # b
}

# for Linf, K and t0
Linf.s = c(); K.s = c(); t0.s = c()

for (n in 1: r)
{
  boot.G = sample(g, size = g, replace = T)

  Linf.s [n] = mean(G[boot.G,1]) # Linf
  K.s [n] = mean(G[boot.G,2]) # K
  t0.s [n] = mean(G[boot.G,3]) # t0
}
```

In the above pseudocodes, **a.s**, **b.s**, **Linf.s**, **K.s**, and **t0.s** are vectors of r random realizations of a , b , L_∞ , K , and t_0 , respectively; # indicates a comment. Consistent with the central limit theorem, the distributions of **a.s**, **b.s**, **Linf.s**, **K.s**, and **t0.s** are approximately normally distributed. Unlike the sampling that would be done separately for each growth parameter, the joint sampling of growth parameters preserves the relationships between the observed data.

The previous resampled values of a , b , L_∞ , K , and t_0 are then subjected to a second, optional step of random draws, from a bivariate normal distribution for a and b and from a trivariate normal distribution for L_∞ , K , and t_0 (depending on species, this step can yield unrealistic random values of a , b , L_∞ , K , or t_0 that can cause crashes of simulations for eqs. 2a–7b):

$$(a, b) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}); \boldsymbol{\mu} = \left\{ \mu_{as} = \frac{1}{r} \sum_{s=1}^r a_s, \mu_{bs} = \frac{1}{r} \sum_{s=1}^r b_s \right\}$$

$$\text{and } \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_a^2 & \text{cov}(a, b) \\ \text{cov}(b, a) & \sigma_b^2 \end{bmatrix}$$

$$(L_\infty, K, t_0) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}); \boldsymbol{\mu} = \left\{ \mu_{L_\infty s} = \frac{1}{r} \sum_{s=1}^r L_{\infty s}, \mu_{Ks} = \frac{1}{r} \sum_{s=1}^r K_s, \mu_{t_0 s} = \frac{1}{r} \sum_{s=1}^r t_{0s} \right\}$$

$$\text{and } \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{L_\infty}^2 & \text{cov}(L_\infty, K) & \text{cov}(L_\infty, t_0) \\ \text{cov}(K, L_\infty) & \sigma_K^2 & \text{cov}(K, t_0) \\ \text{cov}(t_0, L_\infty) & \text{cov}(t_0, K) & \sigma_{t_0}^2 \end{bmatrix}$$

For each type of growth parameter, $\boldsymbol{\mu}$ is a vector of means and $\boldsymbol{\Sigma}$ is a variance–covariance matrix, which must be symmetric positive definite. Sampling is performed with the **mvrnorm** function in the R package MASS (Venables and Ripley 2002). The following are pseudocodes for estimating $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ and for sampling:

```
## for a and b
means.ab = c(mean.a = mean(a.s), mean.b = mean(b.s)) # vector of means

sigma = cov(cbind(a.s, b.s)) # variance-covariance matrix

abdata = mvrnorm(n = r, mu = means.ab, sigma, empirical = TRUE, EISPACK = FALSE)

## for Linf, K, and t0
means.LKt0 = c(mean.Linf = mean(Linf.s), mean.K = mean(K.s), mean.t0 = mean(t0.s)) # vector of means

sigma = cov(cbind(Linf.s, K.s, t0.s)) # variance-covariance matrix

LKt0data = mvrnorm(n = r, mu = means.LKt0, sigma, empirical = TRUE, EISPACK = FALSE)
```

The object **abdata** is a matrix whose column names correspond to a and b ; the object **LKt0data** is a matrix whose column names correspond to L_∞ , K , and t_0 . The parameters a , b , L_∞ , K , and t_0 can be accessed as follows:

```
a.s = abdata[,1]; b.s = abdata[,2] # for a and b

Linf.s = LKt0data[,1]; K.s = LKt0data[,2]; t0.s = LKt0data[,3] # for Linf, K and t0
```

If the second sampling step is used, the objects **a.s**, **b.s**, **Linf.s**, **K.s**, and **t0.s** override the objects of the same names obtained with the first sampling step.

Uncertainty in maximum rate of recruit production (α) is introduced by sampling α from a log-normal distribution error:

$$\alpha \sim \text{LN} \left[\frac{1}{r} \sum_{s=1}^r \ln(e^{jL_{\infty s}^k}), \text{SD} \right]$$

where j and k are, respectively, the intercept and slope of eq. 1; a SD of 0.908 is the SD of residuals for the fitted regression of eq. 1. The pseudocode is as follows:

```
Len.unit = "cm" # Length units in cm; Otherwise "mm"

# "mm"

if (Len.unit == "cm") { # Len.unit is the unit of Linf
  alpha = rlnorm(r, mean = mean(log(exp(15.28)*Linf.s^(-3.344))), sd = 0.908)
} else if (Len.unit == "mm") {alpha = rlnorm(r, mean = mean(log(exp(15.28)*(0.1*Linf.s)^(-3.344))), sd = 0.908)
}
```

Uncertainty in M is introduced by sampling M from a log-normal distribution error:

$$M \sim \text{LN}\left(\frac{1}{r} \sum_{s=1}^r \ln(4.118K_s^{0.73}L_{\infty s}^{-0.33}), \text{SD}\right) \text{ if } L_{\infty} \text{ in cm}$$

or

$$M \sim \text{LN}\left(\frac{1}{r} \sum_{s=1}^r \ln(8.87K_s^{0.73}L_{\infty s}^{-0.33}), \text{SD}\right) \text{ if } L_{\infty} \text{ in mm}$$

```
if (Len.unit == "cm") { # Len.unit is the unit of Linf
M = rlnorm(r, mean = mean(log(4.118*K.s^0.73*Linf.s^-0.33)),
sd = 0.5*mean(4.118*K.s^0.73*Linf.s^-0.33))
}else if (Len.unit == "mm"){M= rlnorm(r, mean = mean(log(8.87*K.s^0.73*Linf.s^-0.33)),
sd = 0.5*mean(8.87*K.s^0.73*Linf.s^-0.33))
}
```

A CV value of 0.5 was suggested by [MacCall \(2009\)](#) as a minimal default value for the variability of constant M on log scale.

Length at age (including L_r , length associated with age-1), natural mortality at age, survivorship to an age, asymptotic weight (kg) and mean weight (kg), the statistic λ_t , the unfished SBPR (kg), the compensation ratio, and steepness are simply derived given the above random draws of a , b , L_{∞} , K , t_0 , α , and M (see Supplementary data for details¹).